Book Recommendation and Influence Analysis

Using Apache Spark & Graph Analytics

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Course: 6240 - Large Scale Parallel Data Processing

Simplified Project Overview – Part 1

Core Objectives:

- Influence Analysis: Which books and users have the most influence in the reading community?
- Personalized Recommendations: What books will a particular user enjoy based on their reading history?
- Community Insights: Understanding age, location, and reading patterns of influential users

Dataset Scale:

1.1 million ratings , **278,858** users , **271,379** books

Problem Statement - Part 2

Solution Approach:

Graph-Based Analysis

- PageRank algorithm applied to user-book network to identify influential entities

Collaborative Filtering

- K-Nearest Neighbors algorithm for personalized book recommendations recommendations

Why Apache Spark?

- Handles massive scale efficiently with distributed computing
- GraphFrames library for advanced graph algorithms
- Seamless integration with AWS EMR for cloud scalability

Dataset Description

Source: Kaggle Books Dataset (~124 MB)

Statistics:

271,000
Books

1.1 Million
Ratings

279,000
Users

Three Core Files:

- **Books.csv**: ISBN, titles, authors, publication years, publishers, image URLs
- **Ratings.csv**: User-book interactions with ratings (1-10 scale)
- **Users.csv**: User demographics including location and age

Sampling & Preprocessing

Data Preparation Steps:

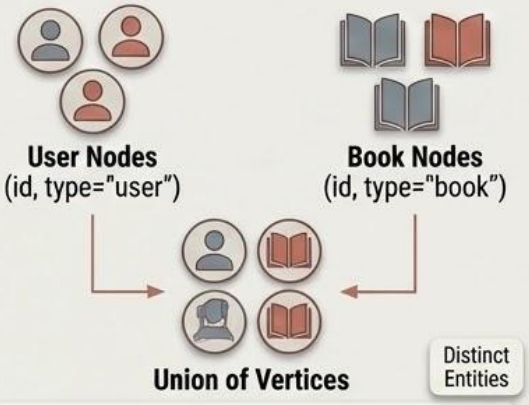
- **Filtering Strategy:**
  Removed books with fewer than 5 ratings to avoid distortion in PageRank results
- **Sampling Approach:**
  Created 30% sample of data for initial testing and validation
- **Performance Optimization:**
  - Implemented Timer class for performance monitoring
  - Normalized column names across datasets
  - Handled CSV separator issues (; instead of ,)
  - Preserved rating distribution in sampled data

Graph Construction & PageRank Execution

Prepare Vertices

Identify Nodes

```
userVertices = select distinct UserID as
               id, "user" as type
bookVertices = select distinct ISBN as id,
               "book" as type
```

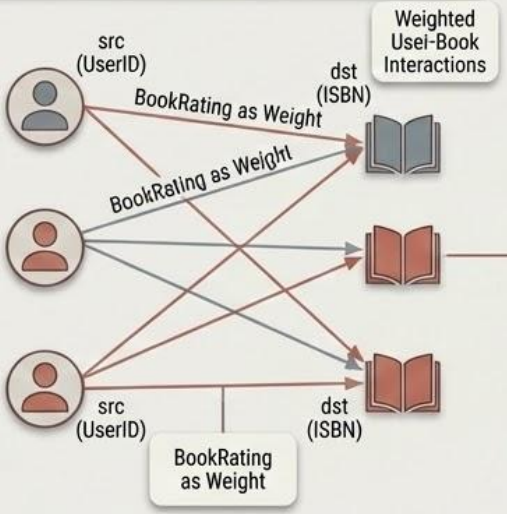



```
vertices = union(userVertices, bookVertices)
```

Prepare Edges

Establish Relationships

```
edges = select UserID as src, ISBN as dst,
        BookRating as weight
```



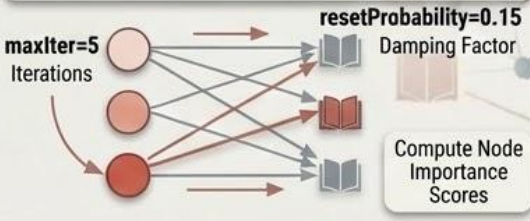
Build Graph & Run PageRank

Create GraphFrame

```
graph = GraphFrame(vertices, edges)
```

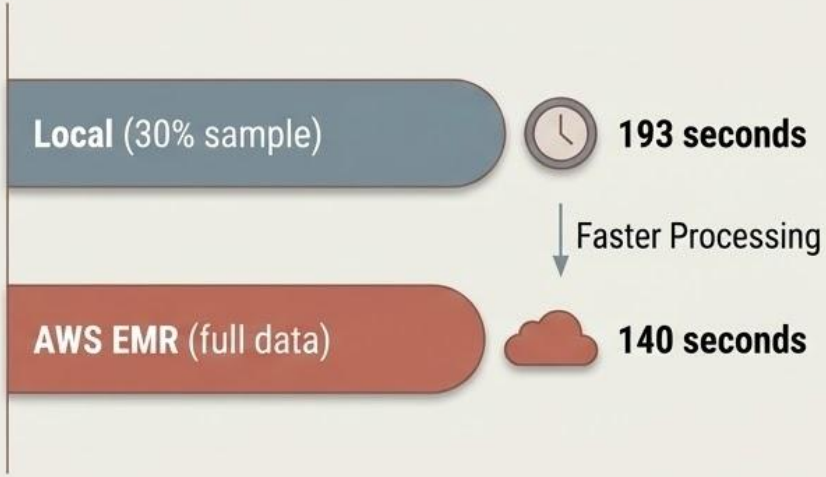
Execute PageRank

```
results = graph.pageRank(maxIter=5,
          resetProbability=0.15)
```



PageRank Results

Performance Metrics



Top Influential Users

User ID	Location	Age
23571	Burbank, California, USA	41
4983	Seattle, Washington, USA	32
3312	Baltimore, Maryland, USA	47
2171	Milltown, Wisconsin, USA	54
957	Sydney, New South Wales, Australia	27

Most influential users aged 25-45, primarily from United States.

K-Nearest Neighbors - Methodology

Recommendation Strategy

1. **Identify Similar Users**
   Find users with similar reading patterns to influential users

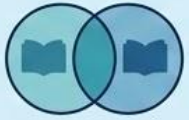
2. **Aggregate Ratings**
   Collect books rated by neighbor users

3. **Filter & Rank**
   Remove already-read books, compute average ratings

4. **Generate Recommendations**
   Select top-rated books with minimum 2 ratings

Quality Thresholds

Minimum 2 ratings per recommended book
Ensures book has sufficient feedback.

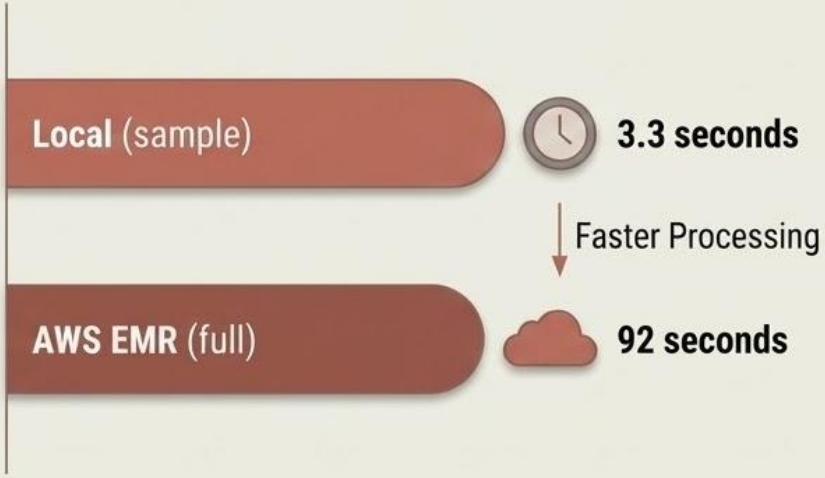


Common books metric for similarity calculation
Based on overlap in reading history.

Average rating prioritization
Higher rated books are preferred.

KNN Results & Recommendations

Performance Metrics



Sample Recommendations

Title	Author	Avg Rating
A Man Named Dave	David J. Pelzer	10.0
Dream a Little Dream	Susan Elizabeth Phillips	10.0
The Black Cauldron	Lloyd Alexander	10.0
Mike Mulligan and His Steam Shovel	Virginia Lee Burton	10.0
A Visit to the Sesame Street Firehouse	Dan Elliott	10.0

Reflection & Learning Outcomes

Key Learnings:

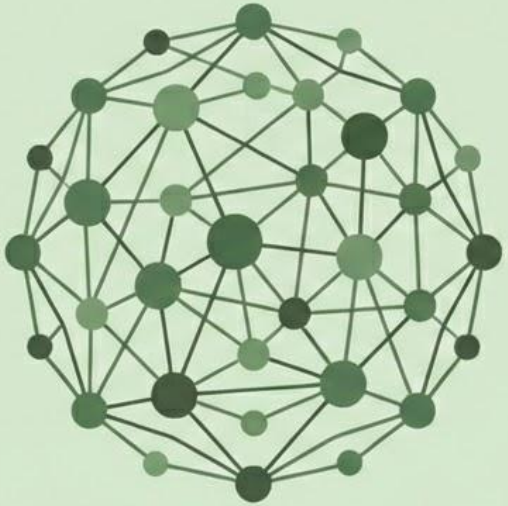
- Spark Mastery: Automated MapReduce operations and parallel processing
- Graph Analytics: Implemented bipartite graph structures for influence analysis
- Algorithm Integration: Connected graph analysis with collaborative filtering
- Cloud Computing: Leveraged AWS EMR for scalable data processing

Technical Skills Developed:

- Large-scale data processing with distributed systems
- Graph-based machine learning algorithms
- Performance optimization and monitoring
- Cloud infrastructure management

Thank You!

Questions?



Contact for more information

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6240 - Large Scale Parallel Data Processing

GitHub | Project Repository